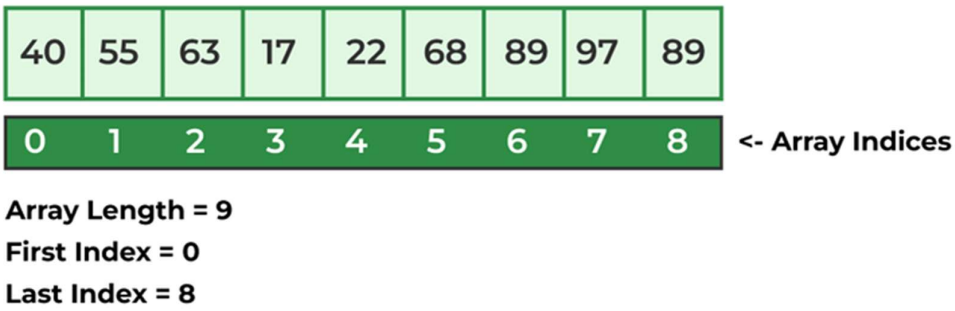
ARRAY IN JAVA

What is Array :-

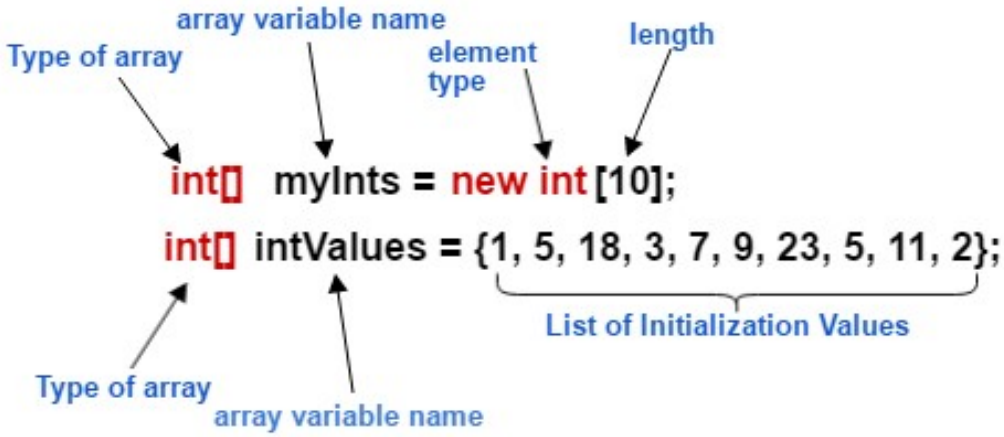
- Array is a **collection of Similar Datatypes.**
- It is a collection of **Homogenous Data.**



Syntax of Array :-

Datatype ArrayName[]=new datatype[Size];

Datatype ArrayName[]={Value1,Value2,Value3…..};

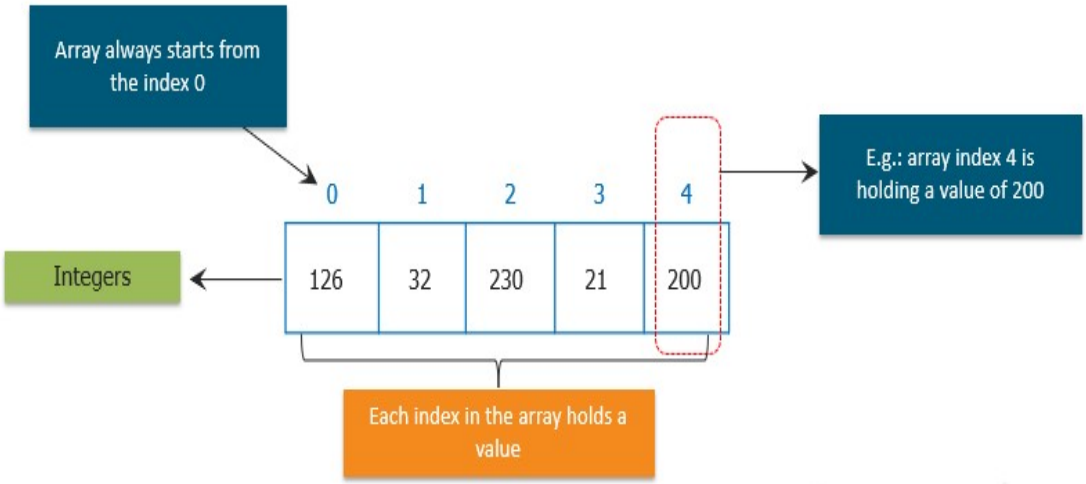


Advantages of Array :-

- Random Access
- Homogenous Data
- We can Manipulate data
- Easy to Update and Retrieve

Disadvantages of Array :-

- Fixed Size
- Can’t store different type of data
- Memory Wastage
- More shifting while insertion and deletion.



Example of Array :-

Input :-

```java
package basicconcepts;

public class arraydemo {
    public static void main(String[] args) {

        int arr[]= {10,20,30,40,50};

        System.out.println("Length of the array is: " +arr.length);

        for(int i=0; i<arr.length; i++)
        {
            System.out.println(arr[i]);
        }
    }
}
```

Output :-

```
Length of the array is: 5
10
20
30
40
50
```

Task 1

Input :-

```java
package assignment;

import java.util.Scanner;

public class arraytask {
    // Print the array of cities entered by user
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the city Size: ");
        int n = sc.nextInt();

        String city[] = new String[n];
        System.out.println("Enter the city names: ");

        for (int i = 0; i < city.length; i++)
        {
            city[i] = sc.next();
        }

        System.out.println("\nList of cities:");
        for (int i = 0; i < city.length; i++)
        {
            System.out.println(city[i]);
        }

    }
}
```

Output :-

```
Enter the city Size: 3
Enter the city names:
Pune Mumbai Solapur

List of cities:
Pune
Mumbai
Solapur
```

Task 2

Input :-

```java
package assignment;

public class arraytask2 {

    //Print the even number of array
    public static void main(String[] args) {

        int arr1[] = {1,2,3,4,5};

        System.out.print("Enter the Number :\n");

        for(int i=0; i<arr1.length; i++)
        {
            if(arr1[i]%2==0)
            {
                System.out.println( arr1[i]);
            }
        }
    }
}
```

Output :-

```
Enter the Number :
2
4
```

Task 3

Input :-

```java
package assignment;

public class arraytask1 {

    // Reverse the array

    public static void main(String[] args) {

        System.out.println("Reverse Number is:");

        int arr1[]= {1,2,3,4,5};

        for(int i=arr1.length-1; i>=0; i--)
        {
            System.out.println(arr1[i]);
        }
    }
}
```

Output :-

```
Reverse Number is:
5
4
3
2
1
```